

## Horticulture at IIT Gandhinagar

IIT Gandhinagar's 400-acre campus is densely planted, and its upkeep produces a large, steady stream of green waste that is processed and reused entirely on site.

### Green Cover

The campus vegetation has three main components:

- **Trees:** more than 1 lakh trees across the campus.
- **Lawns:** large lawn areas used for sports and recreational activities.
- **Hedges:** hedges line the roadsides throughout the campus.

### Planting Concept

The campus planting favours native, low-water, dry deciduous species that suit the arid climate, while retaining most of the mature trees already on site.

**The original site.** Before development, the land was fertile agricultural land, severely eroded into ravines in several places. Large neem and mango trees stood on the flat areas, while ravine thorn forest, with thorny scrub among neem and acacia trees, covered the ravine edges and slopes. An existing herbal garden of neem trees is still part of the campus vegetation network.

**The approach.** The master plan sets the overall landscape framework, and horticulture turns it into a living ecosystem. Key principles:

- Retain most of the mature trees and reinforce the natural vegetation of the ravines.
- Preserve natural lakes, seasonal water bodies and drainage patterns as far as possible.
- Keep development set back from the Sabarmati river to limit impact on its ecosystem.
- Prefer naturalised trees, native shrubs, drought-tolerant plants and groundcovers that need fewer resources.
- Avoid extensive ornamental lawns, which need large quantities of water and maintenance; no new lawn areas are planned in expansion phases.
- Use temporary planting on land kept for future expansion, to prevent barren ground, reduce dust and build green cover gradually.

The tree cover provides shade, improves air quality, moderates the campus microclimate and supports biodiversity.

### Landscape Zones

Each part of the campus has its own planting palette, chosen for its ecological and functional role. The landscape is planned as a connected system of open spaces: entrance drive along the ravines, entrance court and gardens, Central Vista, sports fields, green connectors, River Promenade and the ravine landscape.

| Zone | Planting character |
|------|--------------------|
|------|--------------------|

| Zone                                  | Planting character                                                                                                                                                                                                                                   |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Central Vista                         | Principal open space. Diverse trees, shrubs and groundcovers frame the entrance court, gardens and seating, and create shaded pedestrian avenues linking academic, residential and recreational areas with the water bodies, riverfront and ravines. |
| Circulation routes                    | Rows of broad-canopied avenue trees form the backbone of the green cover, giving a continuous shaded corridor between academic buildings, hostels, residences and recreational areas.                                                                |
| Academic complex, housing and hostels | Medium and small trees, shrubs and groundcovers in courts and arcades. They shade buildings in summer, let sunlight through in winter, and add seasonal colour and fragrance.                                                                        |
| Ravines                               | The ecological heart of the campus. Native dry deciduous trees, shrubs and grasses restore natural vegetation, stabilise slopes, reduce erosion, improve rainwater infiltration and create wildlife habitat.                                         |
| Along the Sabarmati river             | Light, natural planting for soil conservation and erosion control, keeping a sensitive transition between campus and river rather than a heavily landscaped edge.                                                                                    |
| Sports fields                         | Grassed playing fields for sports and recreation.                                                                                                                                                                                                    |

## Tree Species by Zone

The species below are those named for each zone; local names are given where known.

| Zone                           | Species                                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ravines                        | <i>Ziziphus mauritiana</i> (bordi), <i>Acacia nilotica</i> , <i>Acacia leucophloea</i> (harmo), <i>Prosopis cineraria</i> (khijdo), <i>Capparis decidua</i> (kerdo) |
| Ravines and along the boundary | <i>Salvadora oleoides</i> , <i>Acacia auriculiformis</i> , <i>Acacia catechu</i> , <i>Bombax ceiba</i> , <i>Moringa oleifera</i> , <i>Casuarina equisetifolia</i>   |
| Green connecting spaces        | <i>Ailanthus excelsa</i> , <i>Dalbergia sissoo</i> , <i>Butea monosperma</i> , <i>Albizia lebbeck</i> , <i>Madhuca indica</i> ,                                     |

| Zone          | Species                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <i>Mangifera indica</i> , <i>Holoptelea integrifolia</i> ,<br><i>Pongamia pinnata</i> , <i>Manilkara hexandra</i>                                             |
| Central Vista | <i>Tamarindus indica</i> , <i>Syzygium cumini</i> , <i>Azadirachta indica</i> , <i>Ficus amplissima</i> , <i>Ficus benghalensis</i> , <i>Ficus infectoria</i> |

## Irrigation

Landscape irrigation on campus does not rely on fresh water: treated wastewater is the primary source.

**Water sources.** The treated water, including water from root zone treatment systems, is reused for irrigation across the campus, along with non-potable water from the Water Treatment Plant (WTP). As demand has grown, treated wastewater is now getting used completely in horticulture.

### Systems by area.

| Area                             | Irrigation method                    |
|----------------------------------|--------------------------------------|
| Housing courtyards               | Automated drip irrigation            |
| Flat softscape areas             | Automated pop-up sprinklers          |
| Sloping softscape areas          | Automated open 360-degree sprinklers |
| Roadside and pathway plantations | Manual watering where required       |

**Seasons.** Open landscapes are irrigated during cooler hours (night, early morning and evening) to cut evaporation losses.

**Maintenance and expansion.** Filtration at the WTP, before treated water enters the irrigation network, prevents blocked pipes. New housing includes irrigation infrastructure in its landscape design, a proposed third STP will add wastewater recycling capacity, and new hostels and expansion areas will follow native planting with no new lawns.

## Horticulture Team and Upkeep

A horticulture team of over 60 people maintains the campus green cover and keeps it neat and tidy. Hedges and roadside trees are trimmed regularly so they are always kept in shape.

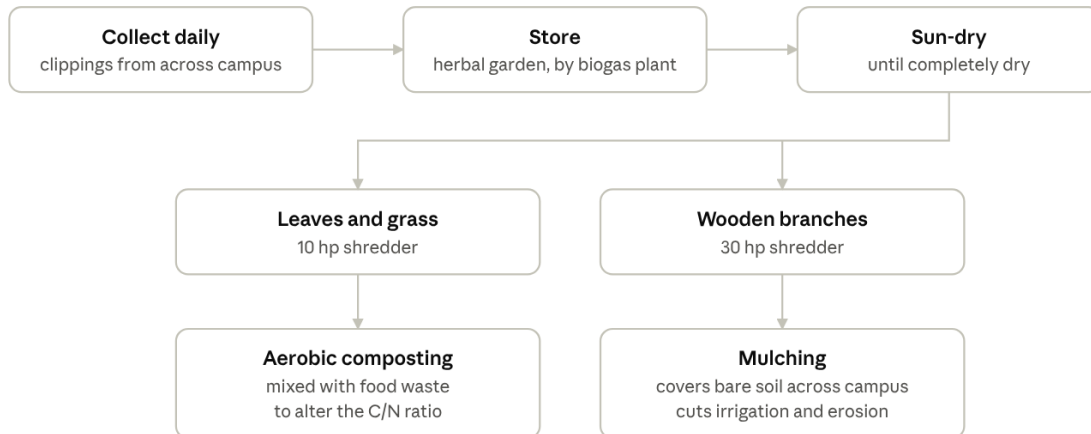
## Horticulture Waste and Daily Collection

Together, the trees, lawns and hedges generate an enormous amount of horticulture waste: leaves, grass clippings and big branches. Clippings from across the campus are collected every day and brought to the herbal garden, right next to the biogas plant, where they are stored at a designated site.

## Drying and Shredding

The clippings are left at the designated site to sun-dry until they are completely dry, which usually takes 2 to 3 months. The dried material is then split into two streams: leaves and grass clippings go through a small 10 hp shredder, and big wooden branches through a 30 hp shredder.

**Leaves go to compost, wood goes to mulch**



## Reuse on Campus

Both shredded streams are reused on campus, so horticulture waste does not leave the site.

- **Leaf and grass shred:** mixed with food waste in aerobic composting to balance the carbon-to-nitrogen (C/N) ratio. Leaves are much richer in carbon than food waste, so they are added in a proportion that keeps the mix at around 30:1.
- **Wood chips:** spread across the campus to cover bare soil, which cuts irrigation demand and reduces soil erosion.